

This is from the textbook chapter **1-2 Parametrized Curve**.

1 Find a parametrized curve $\alpha(t)$ whose trace is the circle

$$x^2 + y^2 = 1$$

such that $\alpha(t)$ runs clockwise around the circle with $\alpha(0) = (0, 1)$.

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2 Let $\alpha(t)$ be a parametrized curve which does not pass through the origin. If $\alpha(t_0)$ is a point of the trace of α closest to the origin and $\alpha'(t_0) \neq 0$, show that the position vector $\alpha(t_0)$ is orthogonal to $\alpha'(t_0)$.

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3 A parametrized curve $\alpha(t)$ has the property that its second derivative $\alpha''(t)$ is identically zero. What can be said about α ?

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4 Let $\alpha: I \rightarrow \mathbb{R}^3$ be a parametrized curve and let $v \in \mathbb{R}^3$ be a fixed vector. Assume that $\alpha'(t)$ is orthogonal to v for all $t \in I$ and that $\alpha(0)$ is also orthogonal to v . Prove that $\alpha(t)$ is orthogonal to v for all $t \in I$.

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5 Let $\alpha: I \rightarrow \mathbb{R}^3$ be a parametrized curve, with $\alpha'(t) \neq 0$ for all $t \in I$. Show that $|\alpha(t)|$ is a nonzero constant if and only if $\alpha(t)$ is orthogonal to $\alpha'(t)$ for all $t \in I$.

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